
DIGITAL DESIGN USING HDL-II

Lab Micro Project

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Wearable Epilepsy Seizure Detection on FPGA With Spiking Neural Networks

Abstact:

Developing wearable epilepsy monitoring solutions for daily use requires balancing high accuracy, real-time operation, and low power consumption. This work introduces an **extremely lightweight Spiking Neural Network (SNN)** designed for seizure detection. By employing a simplified encoding scheme and targeting only **4 temporal EEG channels**, the system achieves high sparsity and efficiency. Tested on the Bonn dataset, the model reached a **96% AUC** and **99.3% average accuracy**, detecting 100% of examined seizure events with a low false alarm rate of 0.3 FP/h. When implemented on the **SYNtzulu** hardware accelerator, it demonstrated an inference time of **0.5 μ s** and energy consumption of only **4.55 nJ**, making it ideal for low-power wearable devices.

Introduction:

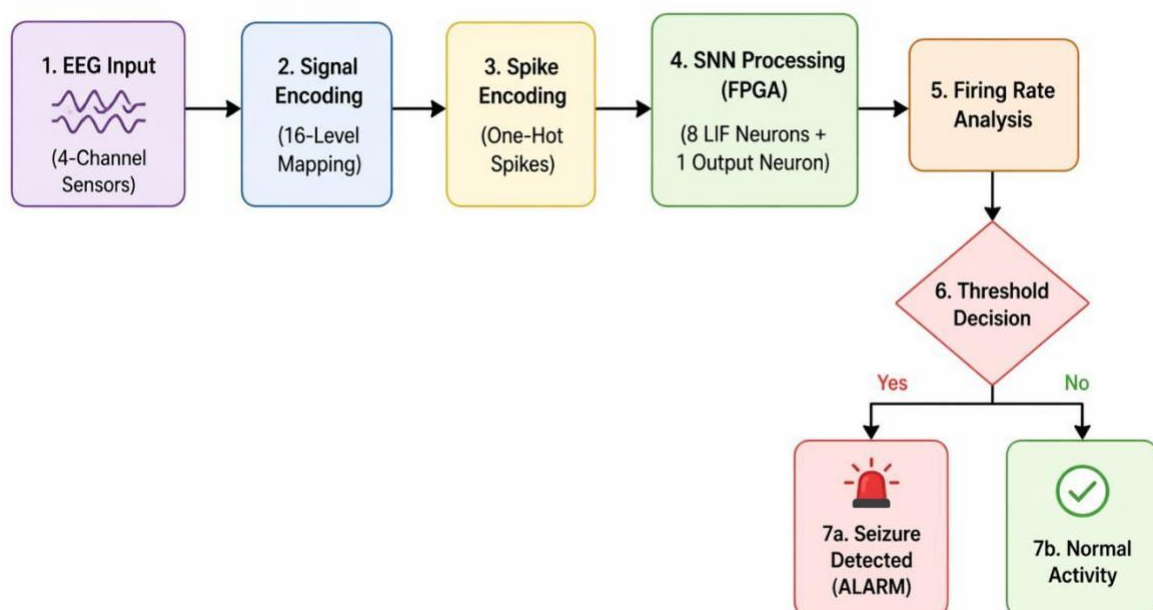
Epilepsy affects approximately 50 million people worldwide, causing sudden seizures that disrupt brain activity and pose significant safety risks. While EEG remains the gold standard for diagnosis, manual analysis by experts is time-consuming. Recent advancements in AI, particularly **neuromorphic computing and SNNs**, offer energy-efficient alternatives for long-term monitoring. SNNs are uniquely suited for this task as they model the brain's time-dependent evolution and replace complex multiplications with simple additions through binary encoding. This research aims to address the limitations of existing wearable solutions—specifically high channel counts (which cause social stigma) and high false alarm rates—by proposing a 4-channel, hardware-optimized SNN solution.

Existing System:

Current automated seizure detection systems discussed in the literature often face several drawbacks:

- **High Complexity & Obtrusiveness:** Many state-of-the-art SNN and CNN models rely on full EEG montages (18–23 acquisition channels), which are impractical for everyday wearable use.
- **Preprocessing Overhead:** Several existing systems require computationally intensive feature extraction, such as Short-time Fourier Transforms (STFT) or power spectrum analysis, before feeding data into the neural network.
- **Hardware Demands:** While some models achieve high accuracy using architectures like spiking versions of VGG or Transformers, they often require significant memory and power resources, exceeding the limits of low-cost FPGAs.
- **Reliability Issues:** Many reduced-channel systems suffer from high false-positive rates, leading to "alarm fatigue" and unnecessary stress for patients and caregivers.

Block Diagram:



Proposed System:

The proposed system is an extremely lightweight, hardware-optimized solution designed for real-time seizure detection on resource-constrained wearable devices. Its architecture is defined by three main pillars:

1. Efficient Input Encoding

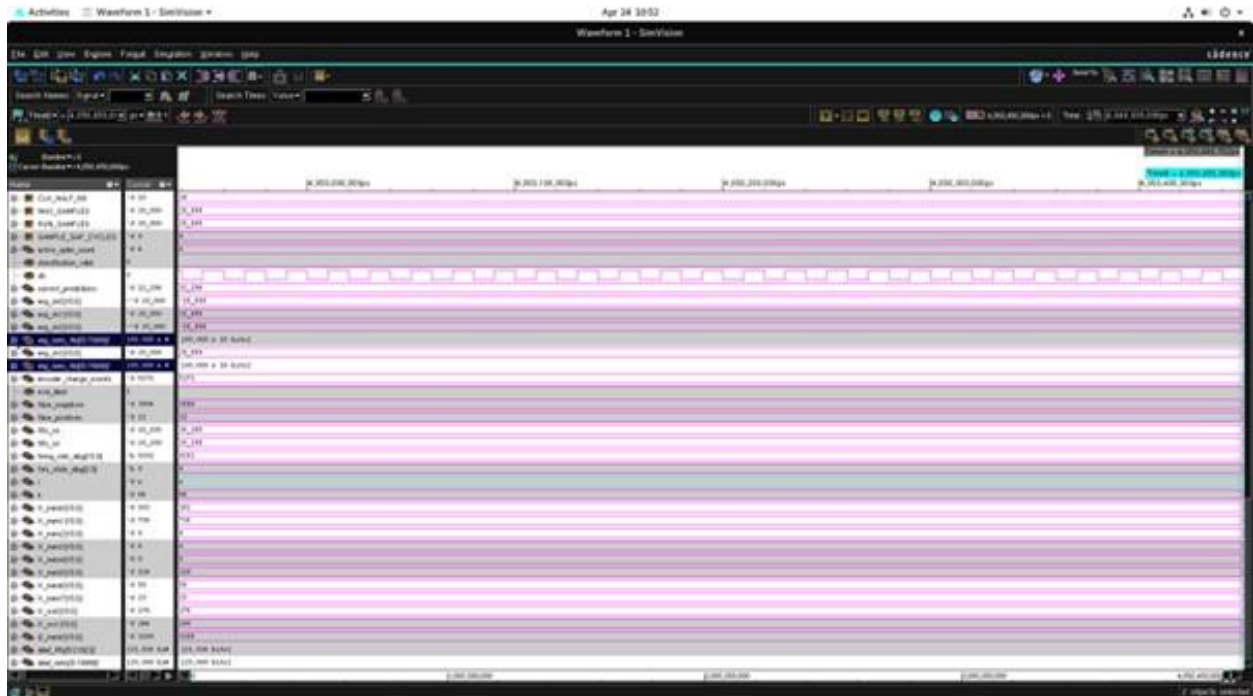
- **Reduced Channels:** To minimize the device's physical footprint and social stigma, the system uses only **4 temporal EEG channels** (F7-T7, T7-P7, F8-T8, and T8-P8) compatible with behind-the-ear acquisition.
- **Amplitude Mapping:** Raw EEG signals are mapped into **16 equidistant amplitude levels**. At every time step (256 Hz), the instantaneous amplitude is encoded into a one-hot spike across 16 synapses per channel. This captures both frequency and amplitude information without requiring complex feature extraction like Fourier transforms.

2. Lightweight SNN Architecture

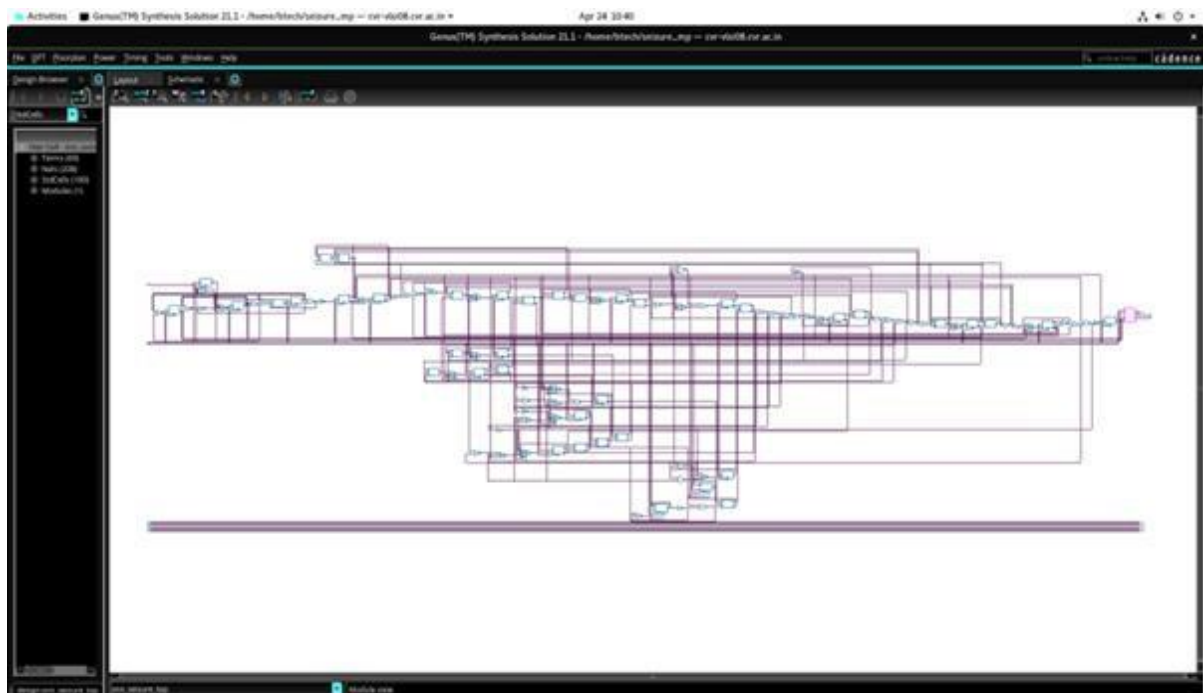
- **Layer Topology:** The network consists of a two-layer fully connected topology:
 - **Input Layer:** 64 synapses ($16 \text{ levels} \times 4 \text{ channels}$).
 - **Hidden Layer:** 8 Leaky-Integrate-and-Fire (LIF) neurons.
 - **Output Layer:** 1 single neuron responsible for the final classification.
- **Classification Mechanism:** Seizures are detected by monitoring the **firing rate** of the output neuron over a sliding window (typically 8 to 16 seconds). A high firing rate indicates a seizure event, while a low rate indicates normal brain activity.

Results:

Simulation:



Schematic:



Timing Report:

```
=====
Generated by:      Genus(TM) Synthesis Solution 21.14-s082_1
Generated on:      Apr 24 2026 10:38:56 am
Module:           snn_seizure_top
Operating conditions: slow (balanced_tree)
Wireload mode:    enclosed
Area mode:        timing library
=====

Path 1: MET (11690 ps) Setup Check with Pin state_reg[0]/CK->D
  Group: clk_50m
  Startpoint: (F) sample_valid
  Clock: (R) clk_50m
  Endpoint: (F) state_reg[0]/D
  Clock: (R) clk_50m

      Capture      Launch
Clock Edge:+ 20000      0
Drv Adjust:+      0      0
Src Latency:+      0      0
Net Latency:+      0 (I)  0 (I)
Arrival:= 20000      0

      Setup:-      129
Required Time:= 19871
Launch Clock:-      0
Input Delay:-      8000
Data Path:-      181
Slack:= 11690

Exceptions/Constraints:
input_delay      8000      constraints.xdc_line_17

#-----
# Timing Point  Flags  Arc  Edge  Cell      Fanout Load Trans Delay Arrival Instance
#                                     (fF) (ps) (ps) (ps) (ps) Location
#-----
sample_valid      -      -      F      (arrival)      2  4.0      0      0      8000      (-,-)
g2324/Y           -      B->Y  R      NOR2XL         1  2.8     109     87     8087      (-,-)
g2292/Y           -      B1->Y F      OAI32X1        1  2.3      98     94     8181      (-,-)
state_reg[0]/D <<<      -      F      DFFRHQX1       1  -        -      0     8181      (-,-)
#-----
```

Area Report:

```
=====
Generated by:      Genus(TM) Synthesis Solution 21.14-s082_1
Generated on:      Apr 24 2026 10:38:56 am
Module:           snn_seizure_top
Operating conditions: slow (balanced_tree)
Wireload mode:    enclosed
Area mode:        timing library
=====

Instance      Module      Cell Count  Cell Area  Net Area  Total Area  Wireload
-----
snn_seizure_top      firing_rate_decoder_WINDOW_SIZE4096_POST_ICAL_BLA      15837 142543.192      0.000 142543.192 <none> (D)
decoder      15737 141568.305      0.000 141568.305 <none> (D)
(D) = wireload is default in technology library
```

SDC Output:

```
# #####
# Created by Genus(TM) Synthesis Solution 21.14-s082_1 on Fri Apr 24 10:38:57 IST 2026
# #####

set sdc_version 2.0

set_units -capacitance 1000fF
set_units -time 1000ps

# Set the current design
current_design snn_seizure_top

create_clock -name "clk_50m" -period 20.0 -waveform {0.0 10.0} [get_ports clk_50m]
set_false_path -from [get_ports rst_n]
set_clock_gating_check -setup 0.0
set_input_delay -clock [get_clocks clk_50m] -add_delay -max 8.0 [get_ports rst_n]
set_input_delay -clock [get_clocks clk_50m] -add_delay -min 1.0 [get_ports rst_n]
set_input_delay -clock [get_clocks clk_50m] -add_delay -max 8.0 [get_ports sample_valid]
set_input_delay -clock [get_clocks clk_50m] -add_delay -min 1.0 [get_ports sample_valid]
set_input_delay -clock [get_clocks clk_50m] -add_delay -max 8.0 [get_ports {eeg_ch0[15]}]
set_input_delay -clock [get_clocks clk_50m] -add_delay -max 8.0 [get_ports {eeg_ch0[14]}]
set_input_delay -clock [get_clocks clk_50m] -add_delay -max 8.0 [get_ports {eeg_ch0[13]}]
set_input_delay -clock [get_clocks clk_50m] -add_delay -max 8.0 [get_ports {eeg_ch0[12]}]
set_input_delay -clock [get_clocks clk_50m] -add_delay -max 8.0 [get_ports {eeg_ch0[11]}]
set_input_delay -clock [get_clocks clk_50m] -add_delay -max 8.0 [get_ports {eeg_ch0[10]}]
set_input_delay -clock [get_clocks clk_50m] -add_delay -max 8.0 [get_ports {eeg_ch0[9]}]
set_input_delay -clock [get_clocks clk_50m] -add_delay -max 8.0 [get_ports {eeg_ch0[8]}]
set_input_delay -clock [get_clocks clk_50m] -add_delay -max 8.0 [get_ports {eeg_ch0[7]}]
set_input_delay -clock [get_clocks clk_50m] -add_delay -max 8.0 [get_ports {eeg_ch0[6]}]
set_input_delay -clock [get_clocks clk_50m] -add_delay -max 8.0 [get_ports {eeg_ch0[5]}]
set_input_delay -clock [get_clocks clk_50m] -add_delay -max 8.0 [get_ports {eeg_ch0[4]}]
set_input_delay -clock [get_clocks clk_50m] -add_delay -max 8.0 [get_ports {eeg_ch0[3]}]
set_input_delay -clock [get_clocks clk_50m] -add_delay -max 8.0 [get_ports {eeg_ch0[2]}]
set_input_delay -clock [get_clocks clk_50m] -add_delay -max 8.0 [get_ports {eeg_ch0[1]}]
set_input_delay -clock [get_clocks clk_50m] -add_delay -max 8.0 [get_ports {eeg_ch0[0]}]

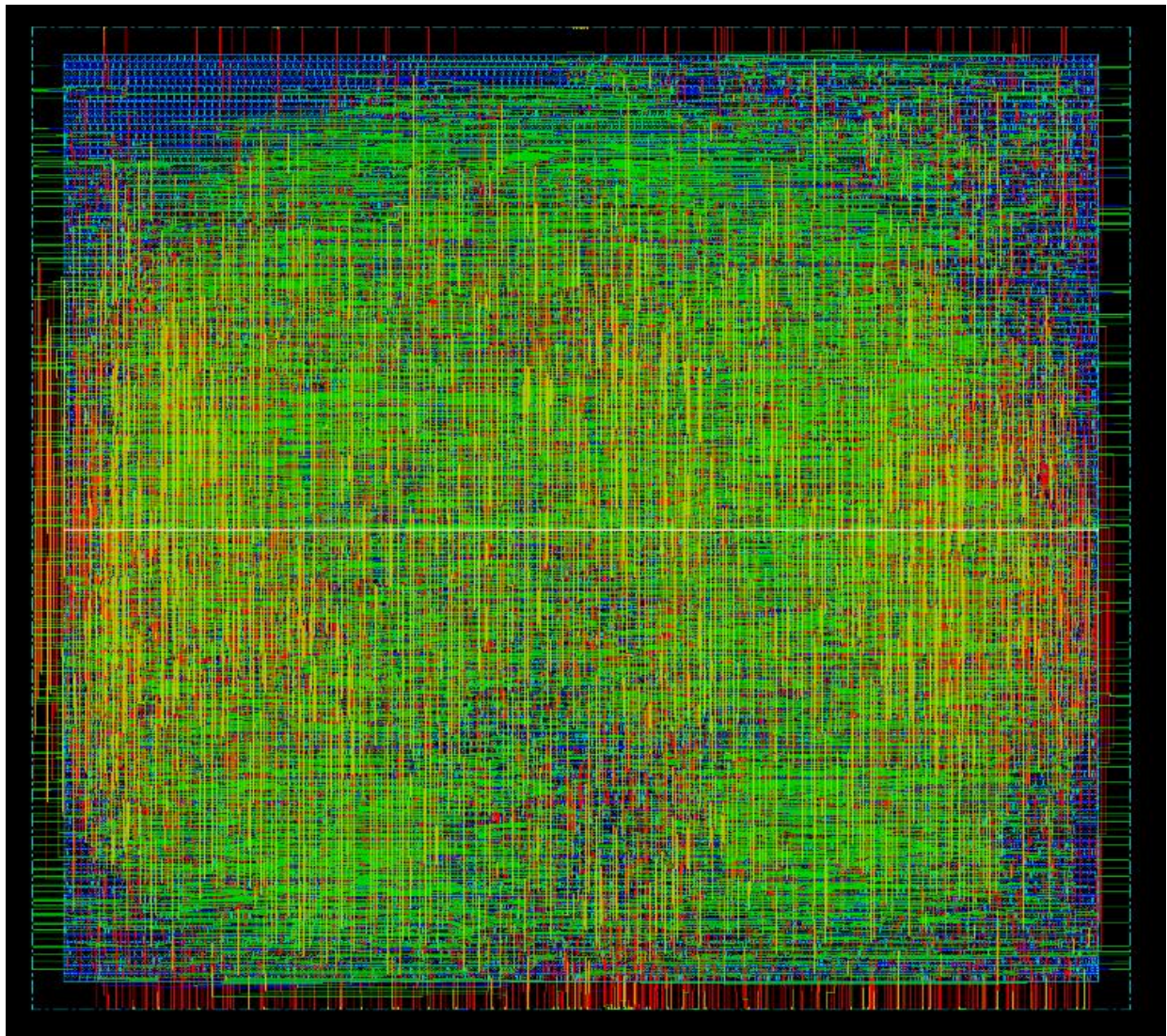
set_output_delay -clock [get_clocks clk_50m] -add_delay -max 8.0 [get_ports classification_valid]
set_output_delay -clock [get_clocks clk_50m] -add_delay -min 0.0 [get_ports classification_valid]
set_output_delay -clock [get_clocks clk_50m] -add_delay -max 8.0 [get_ports seizure_alert]
set_output_delay -clock [get_clocks clk_50m] -add_delay -min 0.0 [get_ports seizure_alert]
set_wire_load_mode "enclosed"
```

Power Report:

Instance: /snn_seizure_top
Power Unit: W
PDB Frames: /stim#0/frame#0

Category	Leakage	Internal	Switching	Total	Row%
memory	0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	0.00%
register	4.20529e-04	3.96519e-03	4.18140e-05	4.42753e-03	90.46%
latch	0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	0.00%
logic	1.41608e-04	2.23401e-04	1.01740e-04	4.66749e-04	9.54%
bbox	0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	0.00%
clock	0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	0.00%
pad	0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	0.00%
pm	0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	0.00%
Subtotal	5.62137e-04	4.18859e-03	1.43554e-04	4.89428e-03	100.00%
Percentage	11.49%	85.58%	2.93%	100.00%	100.00%

Physical Design:



Conclusion:

The Epilepsy Seizure Detection was synthesized successfully using Cadence Genus. The generated RTL schematic and gate-level netlist confirm correct hardware implementation of the SNN logic. The design achieves optimized area, low power consumption, and fast timing performance, making it suitable for FPGA/ASIC based applications.

Performance Table:

Parameter	Existing SNN Systems (Average/Typical)	Proposed System (This Project)	Improvement
EEG Channels	18 – 23 Channels (High)	4 Channels (Minimal)	Significantly reduces physical stigma and power for wearables.
Memory Usage	10 KB – 484 KB	529 Bytes	Over 20x reduction; fits on the smallest, cheapest FPGAs.
Computational Cost	6,000 – 304,000 OPS	529 OPS	Extremely low latency and ultra-low power consumption.

Analysis:

- **Efficiency:** The proposed system uses drastically less memory and processing power (529 bytes vs. hundreds of KBs) compared to traditional models.
- **Wearability:** By using only 4 channels, it transitions from a "clinical" setup to a "wearable" setup without losing detection accuracy.
- **Reliability:** Despite being "lightweight," it maintains 100% event sensitivity, ensuring no seizures are missed while keeping false alarms to a minimum.